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Design and Application of Neural Network for Compensation of VSI Output Voltage Nonlinearities

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Abstract—Voltage source inverters (VSI) with modern power-switching elements are often used to control industrial AC motors. However, the non-linearities of the inverters, such as dead time, turn-on and turn-off switching delay times and voltage drops, are often behind the distortion of the phase currents of the controlled motor. The current distortions can be suppressed by appropriately calculated non-linear functions, which represent the compensation voltages and are consequently added to the control values of the current regulators in the field-oriented control (FOC) algorithm. An artificial neural network (ANN) was designed to identify the non-linear functions of the compensation voltages, which is presented in this paper. Only signals available in the FOC algorithm are used as ANN inputs. The learning process of the neural network takes place online during the running of the motor control algorithm. The learning pattern is generated in each step of the control algorithm from the control errors of the current controllers and the previous ANN outputs. It is not necessary to know the VSI parameters when learning the neural network. The proposed ANN and back-propagation learning algorithm were implemented on one core of the AURIX microcontroller TC397. The proposed strategy was validated through experiments on a real permanent magnet synchronous motor (PMSM), and experimental results prove the effectiveness of the ANN.

Index Terms—Inverter nonlinearities, Artificial neural network, Dead-time compensation, Permanent magnet synchronous motor

I. INTRODUCTION

Voltage Source Inverters (VSI) that use modern power-switching components are frequently used to control industrial AC motors. However, due to the non-linearities of the inverters, such as dead time, turn-on and turn-off switching delays, and voltage drops, there is voltage distortion between the reference voltages and the inverter output voltages. As a result of this phenomenon, phase currents of the controlled motor contain harmonic distortions, which have a negative impact on the applied control algorithm.

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The standard way to reduce output voltage distortion of the VSI is by using the sign compensation method, which relies on accurate polarity detection of the phase currents [1]. This is critical, especially in the region where the phase currents cross zero, as is described in [2], [3]. The papers [4], [5] describe the methods independent of the phase currents polarity detection. The compensation strategy in [5] relies on the injection of the harmonic signals that are calculated based on the mathematical model of the inverter harmonic distortion. However, the effectiveness of this strategy is dependent on the precision of the mathematical model.

The compensation strategy described in the paper [6] utilizes a second-order adaptive filter to extract the 6th harmonic component of the dq - currents. The authors calculated the compensation voltages based on the extracted value of the 6th harmonics and the mathematical model of the permanent magnet synchronous motor (PMSM). However, this approach has a weakness: it does not eliminate higher parasitic harmonic components such as 12th harmonics and other multiples of the 6th harmonics in dq - currents. The paper [7] uses the trapezoidal voltage compensation method that simultaneously controls the angle and amplitude of the trapezoidal voltage. The method is dependent on a precise mathematical description of the inverter's behaviour and an accurate knowledge of the inverter's parameters, such as parasitic capacitances. The authors in the paper [8] used a compensation strategy based on calculating the compensation voltage amplitude based on the voltage differences observed after the injection of the currents.

Authors increasingly use neural networks to suppress voltage distortions of the VSI. In [9], the compensation method is based on the assumption that the vector trajectory of the output $\alpha\beta$ - voltage is a circle. Depending on the value of the circular trajectory error, an adaptation rule is constructed to adjust one shared weight in the input layer of the simple neural network. Subsequently, the neural network can identify the non-linear functions of compensation voltages. The disadvantage is that the neural network adapts only one weight, and others are set to a constant value. In [10], the authors used one separated neuron to suppress 6th and 12th harmonic components in dq - currents. The authors in both articles do not use the full potential of the neural networks to identify the non-linear function of compensation voltages.

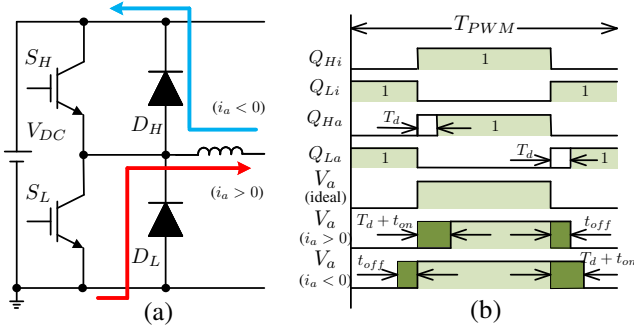


Fig. 1. (a) One leg of the three-phase VSI during the dead-time period. (b) PWM signal patterns and VSI output voltage considering the non-linear effects.

To address the problem of VSI output voltage distortion, an artificial neural network is created to approximate the non-linear functions of the compensation voltages. The signals available in the field-oriented control (FOC) algorithm are used as ANN inputs. The learning process of the ANN is performed online during the PMSM operation. A new learning pattern is generated at every step of the control algorithm. The proposed ANN and learning algorithm is implemented on one core of the microcontroller AURIX TC397. Experiments on a real PMSM validate the effectiveness of the compensation strategy.

II. ANALYSIS OF DISTURBANCE VOLTAGES

The dead-time issue can be explained on the one leg of the typical three-phase VSI shown in Fig.1(a). This dead-time period T_d serves as protection against the simultaneous conduction of the two switching devices on the same leg of the VSI. It is inserted into the gating signals of switching devices. In the dead-time period, the switching elements S_H/S_L on the same leg are turned off, and the output voltage is not controlled. As a result, the phase current can only flow through the diodes D_H/D_L and the actual direction of the phase current determines the output voltage, as shown in Fig.1(b). Even if the dead-time period T_d is very short, there is still a deformation of the VSI output voltage [11].

The voltage error ΔV_{an} can be defined with respect to the switching period T_{PWM} by following equation

$$\Delta V_a = v_{dead} \cdot \text{sign}(i_a) = \frac{T_d + t_{on} - t_{off}}{T_{PWM}} V_{DC} \cdot \text{sign}(i_a) \quad (1)$$

where V_{DC} represents DC bus voltage and the t_{on} and t_{off} represent turn-on/turn-off switching delay time, and the sign function of the a -phase current i_a can be represented as

$$\text{sign}(i_a) = \begin{cases} 1, & i_a > 0 \\ -1, & i_a < 0 \end{cases} \quad (2)$$

The equation for calculating the compensation voltage v_{dead} is frequently expressed in the form (3), for example in [1]

$$v_{dead} = \frac{T_d + t_{on} - t_{off}}{T_{PWM}} (V_{DC} - V_{sat} + V_d) + \frac{V_{sat} + V_d}{2} \quad (3)$$

where V_{sat} and V_d denote voltage drops on switching devices and diodes.

The voltage error ΔV_{abc} can be approximated by expansion into the Fourier series and after transformation into the dq -coordinate system expressed according to relation (4)-(5). A detailed description of the harmonic analysis can be found in the [6]. The electric position of the PMSM is represented by θ_e . The γ is the angle between the q -axis of the dq -coordinate system and the current vector.

$$\Delta v_d = \frac{4}{\pi} v_{dead} \left\{ \sin(\gamma) + \sum_{n=1}^{+\infty} \frac{\sin(6n(\theta_e + \gamma) - \gamma)}{6n - 1} + \sum_{n=1}^{+\infty} \frac{\sin(6n(\theta_e + \gamma) + \gamma)}{6n + 1} \right\} \quad (4)$$

$$\Delta v_q = \frac{4}{\pi} v_{dead} \left\{ -\cos(\gamma) + \sum_{n=1}^{+\infty} \frac{\cos(6n(\theta_e + \gamma) - \gamma)}{6n - 1} - \sum_{n=1}^{+\infty} \frac{\cos(6n(\theta_e + \gamma) + \gamma)}{6n + 1} \right\} \quad (5)$$

for $n = 1, 2, 3, 4, \dots$, where $n = 1$ corresponds to the 6th voltage harmonic component. These disturbance voltages cause parasitic harmonic components that appear in the dq -currents at the same frequencies.

III. DESIGN OF COMPENSATION STRATEGY

The analysis of the impact of the dead-time and other VSI non-linearities reveals that disturbance voltages are non-linear functions that depend on the polarity of the phase currents. This observation motivates us to use an artificial neural network (ANN) to address this issue.

Several requirements have been taken into account when designing a suitable neural network.

- Capability to accurately approximate the compensation voltages
- Input signals must be available in the FOC algorithm
- Capability of online learning and compensating during the learning phase
- Suitable for implementation on microcontroller

The performance of the ANN in solving complex problems depends on the number of hidden layers and neurons. Increasing the number of neurons in the layers can help solve more complicated issues, but with added computational complexity.

A. Topology of artificial neural network

The goal is to approximate the simplest form of compensation voltages. It is achievable in dq -coordinates. The basic shape of the compensating voltages in the d - and q -components repeat six times in one electrical revolution. The mathematical description of the compensation voltages in dq -coordinates is given in equation (4)-(5). The neural network topology is chosen as simply as possible with an input layer, two hidden layers and an output layer, as shown in Fig. 2. The hidden layers contain 10 neurons, and the output layer has 2 neurons. The selected topology should have enough degrees of freedom to learn and approximate the desired non-linear

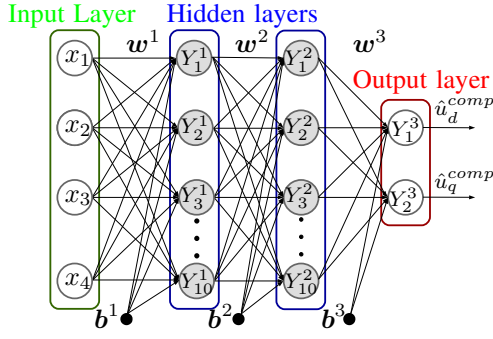


Fig. 2. Topology of the proposed ANN.

signals. In general, the l^{th} layer of the ANN can be described by the following equation

$$\mathbf{y}^{(l)} = \mathbf{f} \left(\mathbf{w}^{(l)} \mathbf{y}^{(l-1)} + \mathbf{b}^{(l)} \right) \quad (6)$$

where $\mathbf{y}^{(l)}$ represents the output of the l^{th} layer and $\mathbf{y}^{(l-1)}$ represents the inputs from the previous layer. The symbol $\mathbf{w}^{(l)}$ denotes the matrix of neuron weights, and $\mathbf{b}^{(l)}$ represents the vector of neuron bias values.

The next essential step for the ANN's proper functionality is choosing suitable input signals that contain important information for extracting the desired compensation voltages. The following equation gives the input vector $\mathbf{x}_i$.

$$\mathbf{x}_i = [\sin(6\theta_e), \cos(6\theta_e), \gamma, I_s^{amp}]^T \quad (7)$$

The input vector contains the $\sin(6\theta_e)$ and $\cos(6\theta_e)$ signals, where $6\theta_e$ represents six times the electrical position. These signals can be used to transform into a new coordinate system related to the rotation angle $6\theta_e$. In this coordinate system, the 6th harmonic component contained in the dq -currents would have a constant value during the steady state (constant speed and load torque of the motor) (Note: The frequency of the 6th harmonic component of the dq -currents is related to the frequency of the fundamental harmonic component of the phase currents.).

The following input is γ angle. Assuming that the d -component of the current is controlled to zero and the PMSM operates in a steady state, the γ angle should be zero in this case (the field weakening is not assumed). Due to the non-linearities of the inverter, the γ angle does not have a zero value. However, this non-zero value contains information about the persistent harmonic distortion, which ANN can use to suppress these distortions.

$$I_s^{amp} = \sqrt{i_a^2 + i_b^2 + i_c^2} \quad (8)$$

The last input signal I_s^{amp} is the magnitude of the current vector in abc -coordinates, which is described by (8). This input information should help the neural network to extract the compensating voltages during transient states of the PMSM.

Determining appropriate activation functions $f(x)$ is a crucial part of the ANN design process. We have chosen the rectified linear unit (ReLU) function for the input and hidden layer as the activation function $f_{relu}(x)$ due to its simplicity

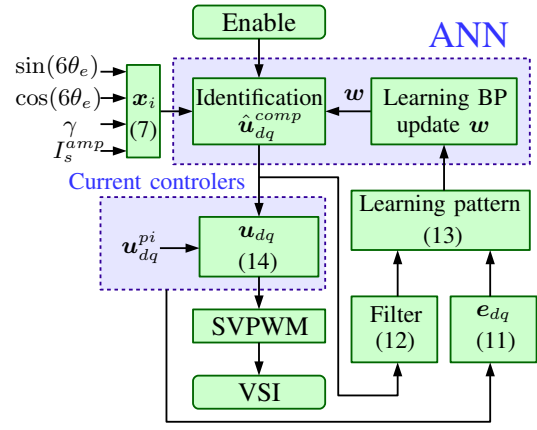


Fig. 3. The flow of the compensation voltages identification process and ANN learning.

in calculating its derivative. Another benefit is that the ReLU function and its derivative are monotonic functions.

$$f_{relu}(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases} \quad f'_{relu}(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases} \quad (9)$$

We chose the linear function as the activation function $f_{out}(x)$ for the output layer.

$$f_{out}(x) = x \quad f'_{out}(x) = 1 \quad (10)$$

B. Online neural network learning process

The learning process of the neural network is designed to train the ANN online directly while the PMSM is under operation conditions. This approach requires assembling a training pattern from the current signal values at each learning process step. Due to its low computational complexity, we used a modified back-propagation (BP) learning algorithm for ANN. The identification process of the compensation voltages and the learning proposed ANN are shown in Fig. 3. The learning pattern $\mathbf{T}^{tp}$ is created by the sum of the previous ANN outputs $\hat{\mathbf{u}}_{dq}^{comp}(k)$ with the vector of errors $\mathbf{e}_{dq}(k)$, representing persistent harmonic distortion in the dq -currents. The following equation gives the vector of the errors.

$$\begin{bmatrix} e_d(k) \\ e_q(k) \end{bmatrix} = \mathbf{K}_{gain} \left(\begin{bmatrix} i_d^{ref}(k) \\ i_q^{ref}(k) \end{bmatrix} - \begin{bmatrix} i_d(k) \\ i_q(k) \end{bmatrix} \right) \quad (11)$$

where $i_d^{ref}(k)$ and $i_q^{ref}(k)$ represent setpoint values of the dq -currents, and $i_d(k)$ and $i_q(k)$ are measured currents. The dq -current errors are multiplied by the correction gain $\mathbf{K}_{gain}$, which corresponds to the R_s stator resistance.

The filter $F_{lpf}(q^{-1})$ filters the output compensation voltages $\hat{\mathbf{u}}_{dq}^{comp}(k-2)$ from the ANN in the previous step to remove the DC component. The symbol q^{-1} denotes the backward shift operator.

$$F_{lpf}(q^{-1}) = 1 - K_{lpf} \frac{bq^{-1}}{1 - aq^{-1}} \quad (12)$$

Due to feedback filtration, the neural network learns to approximate only harmonic ripples that correspond to the mathematical description according to equations (4)-(5) without terms $\sin(\gamma)$ and $-\cos(\gamma)$ which represent DC components in compensation voltages in the steady state.

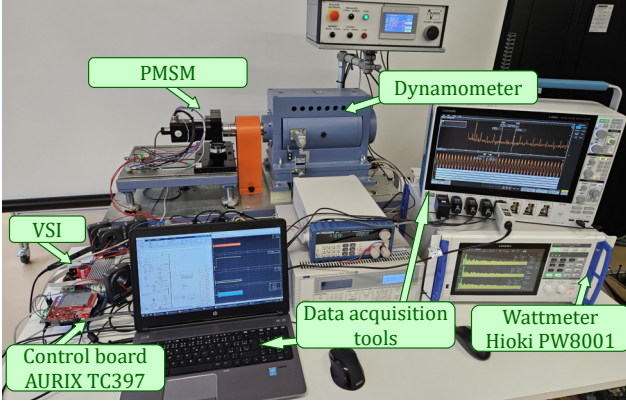


Fig. 4. The configuration of the testing platform.

The training pattern $\mathbf{T}^{tp}(k) = [u_d^{tp}(k), u_q^{tp}(k)]^T$ for the current step of the learning process is calculated according to the following equation.

$$\begin{bmatrix} u_d^{tp}(k) \\ u_q^{tp}(k) \end{bmatrix} = \begin{bmatrix} e_d(k) \\ e_q(k) \end{bmatrix} + \mathbf{F}_{lpf}(q^{-1}) \begin{bmatrix} \hat{u}_d^{comp}(k-2) \\ \hat{u}_q^{comp}(k-2) \end{bmatrix} \quad (13)$$

The compensation voltages identified by the neural network can reach maximum values that are limited by saturation at the output. The limits are set based on the VSI parameters and the expected PMSM operating conditions.

$$\begin{bmatrix} u_d(k) \\ u_q(k) \end{bmatrix} = \begin{bmatrix} u_d^{pi}(k) \\ u_q^{pi}(k) \end{bmatrix} + \begin{bmatrix} u_d^{decoup}(k) \\ u_q^{decoup}(k) \end{bmatrix} + \begin{bmatrix} \hat{u}_d^{comp}(k) \\ \hat{u}_q^{comp}(k) \end{bmatrix} \quad (14)$$

The compensation voltages add to the $u_{dq}^{pi}(k)$ voltages generated by the PI current controllers and decoupling voltages $u_{dq}^{decoup}(k)$.

IV. EXPERIMENTAL RESULTS

The compensation strategy was verified on the test bench, which is shown in Fig. 4. The test bench includes a three-phase PMSM connected to the dynamometer. The control platform with the microcontroller Tricore AURIX TC397 has six separate cores; however, only two are utilized by the application. The first is dedicated to the FOC algorithm and the proposed ANN compensation strategy, including the back-propagation learning algorithm. The sampling frequency of the first core is 10 kHz. The second core is used for data acquisition and communication via UDP with MATLAB/Simulink. The dead-time is set to $2 \mu s$. The other parameters of the VSI with insulated-gate bipolar transistors (IGBT) and parameters of the PMSM are shown in Tab. II.

The experimental setup configuration is as follows: the dynamometer is set as a speed reference $\omega_r^{ref} = 200$ rpm. The PMSM operates in torque mode with a q -current setpoint $i_q^{ref} = 1$ A. This configuration is chosen because the nonlinearities of the inverter are more evident at low engine speeds and low motor loads. At the start of the experiment, all weights and biases of the ANN are initialized to their initial values. The first and second hidden layers have set weights randomly in the range of $\langle -0.5, 0.5 \rangle$ and $\langle -0.2, 0.2 \rangle$ and the output

layer in $\langle -0.1, 0.1 \rangle$. The learning coefficient α has a constant value of 0.04.

The online learning process of the ANN was enabled at 24 seconds of the experiment, as shown in Fig. 6 (b). The identification process of the compensation voltages Δv_{dq} is shown in Fig. 6 (c). The course of d -current during the learning phase is shown in Fig. 6 (b). The ripple that is noticeable primarily in the d -component of the compensation voltage is caused by the transient state of the low-pass filter (LPF). This transient state depends on the selection of the LPF time constant T_{lpf} and gain K_{lpf} . We have chosen $T_{lpf} = 1$ s and $K_{lpf} = 0.05$ for this experiment.

The criterion C_{id} is defined to evaluate the progress of the ANN learning process and the current effectiveness of the compensation. C_{id} indicates the magnitude of the harmonic ripple of the d -current in a steady state, assuming the d -component is controlled to zero.

$$C_{id} = \frac{1}{N} \sum_{k=1}^N |i_d(k)| \quad (15)$$

where i_d represents the measured value of the d -current, and N is the number of samples per electric revolution. C_{id} converged to a constant value (Fig. 6 (b)) and did not further decrease significantly. The residual value of the criterion is mainly determined by the noise contained in the d -component of the current. The transient in the output compensation voltages disappears two seconds after the ANN starts learning. The results obtained after completing the learning phase of the ANN are shown in Fig. 5 (g)-(i) and Fig. 7 (b).

These results are compared with cases without compensation and with the standard compensation method, which uses the average size of the voltage error per duty cycle and the polarity of the phase currents to determine the compensation voltages according to equation (3). The course of the phase currents in Fig. 5 (a) without using any compensation method is not purely sinusoidal because parasitic harmonic components deform them. Other deformations are in the regions where the phase currents pass through the zero value, where the zero-clamping phenomenon is applied. These deformations are transferred via transformations (Clarke, Park) to the dq -coordinate system linked to the electrical position. These deformations cause ripples with the dominant 6^{th} harmonics, which are mainly visible in the i_d current in Fig. 5 (b).

The standard sign compensation particularly suppresses the harmonic deformation of the phase currents and dq -currents, as can be seen in Fig. 5 (d)-(f). However, the harmonic distortion is not completely suppressed, mainly due to inaccurate polarity detection of the phase currents. This is especially noticeable in the d -component, where the current peaks occur in the regions where the phase currents pass through zero. In contrast, the proposed approach with ANN suppresses all harmonic distortion of phase currents and dq -currents significantly, as seen in Fig. 5 (g)-(i). The input signals to the ANN do not include phase currents or their sign functions. The extracted compensation voltages are not affected by them and can effec-

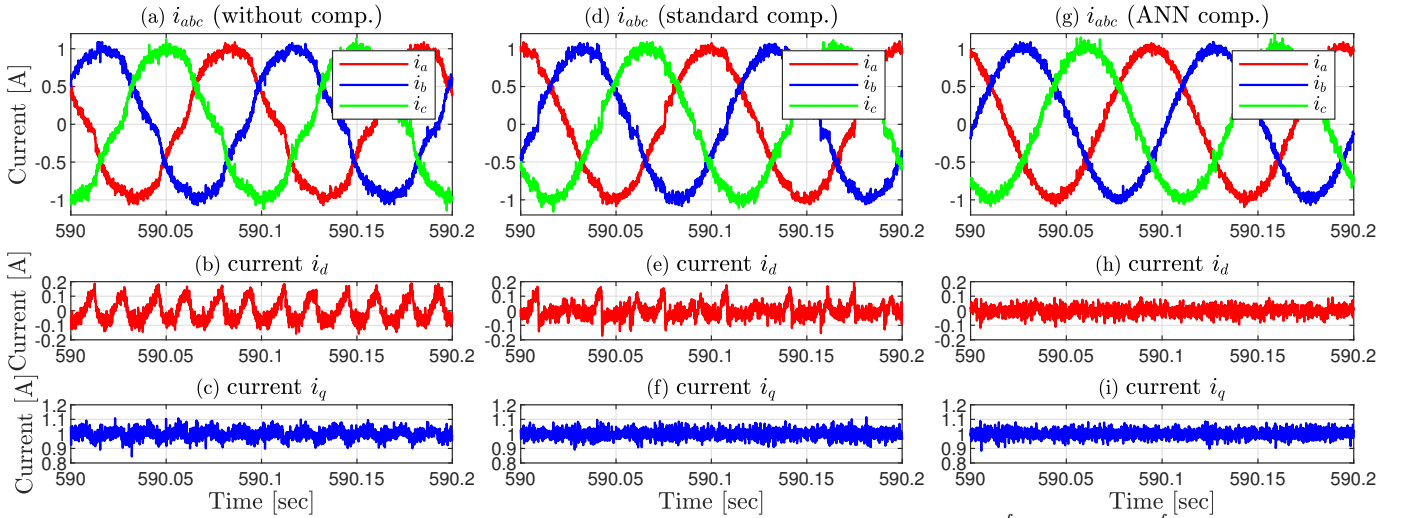


Fig. 5. Phase and dq - currents without/with standard comp. and with proposed ANN comp. ($\omega_r^{ref} = 200$ rpm, $i_q^{ref} = 1$ A).

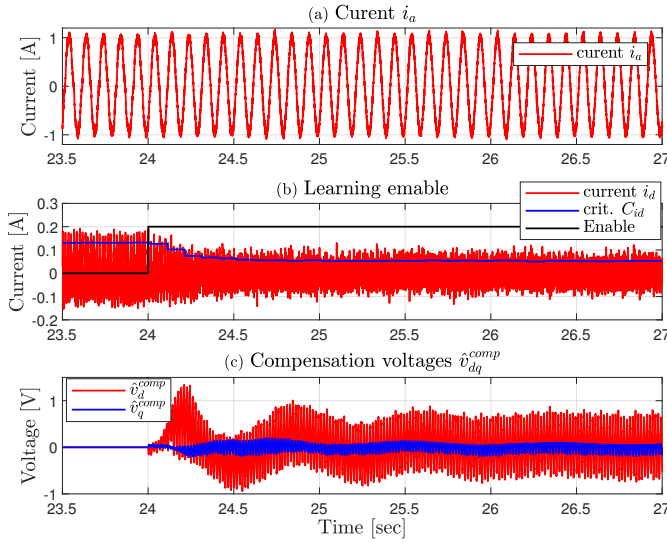


Fig. 6. The online learning process of the proposed ANN on real PMSM.

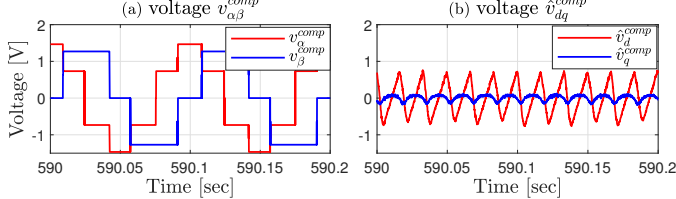


Fig. 7. Compensation voltages from standard compensation method and proposed ANN ($\omega_r^{ref} = 200$ rpm, $i_q^{ref} = 1$ A).

tively eliminate harmonic distortion, even in the areas where the phase currents pass through zero. The XY plot waveforms of currents are shown in Fig. 9 (c). The trajectory of the current vector has a pure circular shape due to the proposed ANN compensation. The details of the compensation voltages from the ANN and the standard compensation strategy in $\alpha\beta$ -coordinates are shown in Fig. 7.

Fig. 8 compares the a -phase currents and dq -currents spectra obtained using fast Fourier transform (FFT) from measured currents in MATLAB. The harmonic distortion of the phase currents is reflected by odd parasitic harmonic components

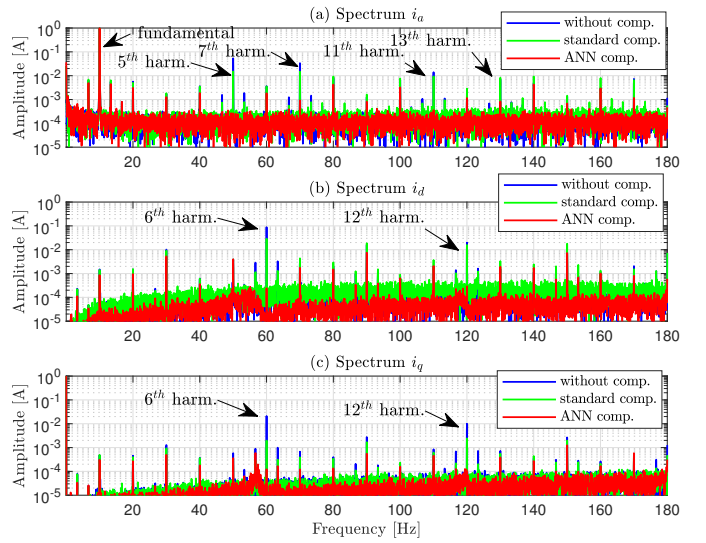


Fig. 8. Spectra of the a -phase and dq - currents without/with standard comp. and with proposed ANN comp. ($\omega_r^{ref} = 200$ rpm, $i_q^{ref} = 1$ A).

contained in spectra, with the 5th, 7th, 11th, and 13th being the dominant harmonics. In the dq -coordinate system, the parasitic harmonic components transform into dominant 6th and 12th harmonic components. Both compensatory strategies suppress these parasitic harmonic components. The standard compensatory strategy only partially suppresses them, but the proposed strategy almost wholly. These results are illustrated in Fig. 10 and Fig. 11, which compares the dominant parasitic harmonic components in the a -phase currents and dq -currents.

Another proof that the proposed compensation strategy achieved significantly better results is the total harmonic distortion (THD) of the a -phase currents measured with the wattmeter Hioki PW8001. The measured values for four operating conditions are listed in Tab. I. The computational complexity of the FOC algorithm is 7 μ s on average, with an additional 5 μ s spent measuring phase currents and generating PWM signals. The ANN learning phase consumes most resources (29 μ s) while the inference only takes 9 μ s on average.

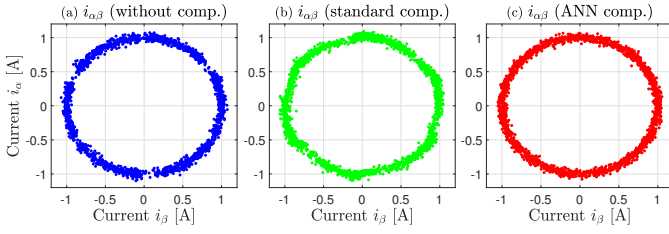


Fig. 9. The XY plot of the $\alpha\beta$ -currents without/with standard comp. and with proposed ANN comp. ($\omega_r^{ref} = 200$ rpm, $i_q^{ref} = 1$ A).

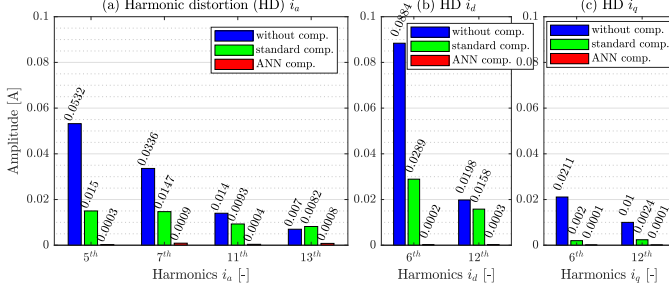


Fig. 10. Comparison of the dominant parasitic harmonic components of the α -phase currents and dq -currents ($\omega_r^{ref} = 200$ rpm, $i_q^{ref} = 1$ A).

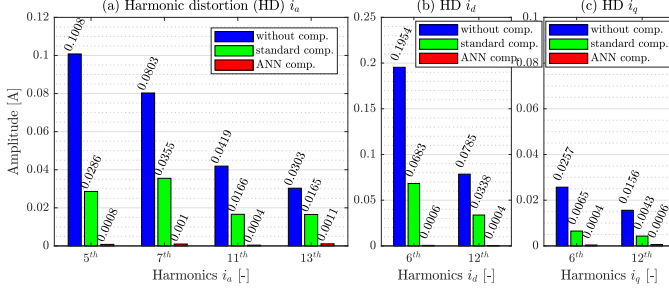


Fig. 11. Comparison of the dominant parasitic harmonic components of the α -phase currents and dq -currents ($\omega_r^{ref} = 400$ rpm, $i_q^{ref} = 4$ A).

TABLE I
COMPARISON OF THD α -PHASE CURRENT

Compensation method	THD [%]			
	200 rpm load 1 A	200 rpm load 4 A	400 rpm load 1 A	400 rpm load 4 A
Without comp.	7.276	2.832	8.166	3.952
Standard comp.	3.842	1.326	5.473	1.811
ANN comp.	2.202	0.637	3.146	0.884

V. CONCLUSION

The paper addresses the challenge of mitigating distortion in the inverter output voltages by proposing a compensation strategy based on an artificial neural network. The proposed ANN topology is designed to provide the balance between the flexibility needed to approximate non-linear compensation voltages and suitability for integration into the Aurix TC397 microcontroller. The effectiveness of the proposed strategy is demonstrated on a test bench featuring PMSM controlled by the FOC algorithm. The learning algorithm can learn ANN online directly on a microcontroller. The output compensation voltages can be applied during the learning process, which is one of its most significant advantages. Results demonstrate

the superior ability of the proposed strategy to suppress undesirable effects of the dead-time and other non-linearities of the inverter. The performance of the proposed strategy is significantly better than standard methods in mitigating current harmonic distortion. The benefit of this solution is independence from additional hardware requirements, but it places higher demands on the computing performance of the microcontroller.

TABLE II
PARAMETERS OF THE PMSM AND VSI

Name	Symbol	Value	Unit
DC voltage	V_{DC}	50	V
Maximum motor current	i_{max}	8.55	A
Back-EMF constant	λ_m	0.0299	Vs/rad
d-axis inductance	L_d	430	μ H
q-axis inductance	L_q	450	μ H
Nominal speed	ω_N	1500	rpm
Maximum speed	ω_{max}	12000	rpm
Nominal power	P_N	180	W
Number of pole pairs	p_p	3	-
Switching device voltage drop	V_{sat}	1.5	V
Diode voltage drop	V_d	1.7	V

REFERENCES

- [1] Sam-Young Kim, Wootaik Lee, Min-Sik Rho, and Seung-Yub Park, "Effective Dead-Time Compensation Using a Simple Vectorial Disturbance Estimator in PMSM Drives," *IEEE Transactions on Industrial Electronics*, pp. 1609–1614, 2010.
- [2] A. S. N and K. Vasudevan, "An Investigation of Current Ripple Prediction Based Dead Time Compensation Using a Simplified Motor Model for a VSI fed Induction Motor Drive," *2023 IEEE International Conference on Power Electronics, Smart Grid, and Renewable Energy (PESGRE)*, pp. 1–6, 2024.
- [3] S. M. Seyyedzadeh and A. Shoulaie, "Accurate Modeling of the Nonlinear Characteristic of a Voltage Source Inverter for Better Performance in Near Zero Currents," *IEEE Transactions on Industrial Electronics*, vol. 66, no. 1, pp. 71–78, jan 2019.
- [4] J. Ye, Y. Huang, S. Huang, H. Wang, B. Li, J. Xu, and A. Shen, "An Accurate Dead Time Compensation Method for SPWM Voltage Source Inverters," *IEEE Transactions on Power Electronics*, vol. 38, no. 4, pp. 4894–4908, apr 2023.
- [5] G. Liu, D. Wang, Y. Jin, M. Wang, and P. Zhang, "Current-Detection-Independent Dead-Time Compensation Method Based on Terminal Voltage A/D Conversion for PWM VSI," *IEEE Transactions on Industrial Electronics*, pp. 7689–7699, oct 2017.
- [6] Z. Wu, H. Zhang, K. Liu, W. Hua, G. Zhang, B. Wang, S. Ding, and J. Hang, "Dead-Time Compensation Based on a Modified Multiple Complex Coefficient Filter for Permanent Magnet Synchronous Machine Drives," *IEEE Transactions on Power Electronics*, vol. 36, no. 11, pp. 12979–12989, nov 2021.
- [7] L. Wang, J. Xu, Q. Chen, Z. Chen, and R. Huang, "An Improved Trapezoidal Voltage Method for Dead-Time Compensation in Three-Phase Voltage Source Converter," *IEEE Transactions on Power Electronics*, vol. 37, no. 8, pp. 8785–8789, 2022.
- [8] K. Yu and Z. Wang, "An Online Compensation Method of VSI Non-linearity for Dual Three-Phase PMSM Drives Using Current Injection," *IEEE Transactions on Power Electronics*, vol. 37, no. 4, pp. 3769–3774, 2022.
- [9] T. Liu, Q. Li, Q. Tong, Q. Zhang, and K. Liu, "An Adaptive Strategy to Compensate Nonlinear Effects of Voltage Source Inverters Based on Artificial Neural Networks," *IEEE Access*, vol. 8, pp. 129992–130002, 2020.
- [10] X. Wang and H. Zhu, "Vibration Compensation Control of BPMSM with Dead-Time Effect Based on Adaptive Neural Network Band-Pass Filter," *IEEE Transactions on Power Electronics*, vol. 37, no. 6, pp. 7145–7155, jun 2022.
- [11] L. Buchta and O. Bartik, "Dead-time compensation strategies based on kalman filter algorithm for pmsm drives," in *IECON 2019 - 45th Annual Conference of the IEEE Industrial Electronics Society*, vol. 1, 2019, pp. 986–991.